

MT8002 Elements of Set Theory
The Zermelo–Fraenkel Axioms
Axioms 4 to 6

Andrés Sicard-Ramírez

Universidad EAFIT

Semester 2026-2

Preliminaries

Textbook

Derek Goldrei (1996). Classic Set Theory. A Guided Independent Study.

Conventions

The numbers and page numbers assigned to chapters, examples, exercises, figures, quotes, sections and theorems on these slides correspond to the numbers assigned in the textbook.

See the appendix in the course homepage for the logical symbols, mathematical symbols and acronyms not defined in these slides.

Outline

Introduction

The Axiom of Separation

The Axiom of Power Set

The Axiom of Union

Intersection of Sets

Cartesian Product

Other Definitions

References

Outline

Introduction

The Axiom of Separation

The Axiom of Power Set

The Axiom of Union

Intersection of Sets

Cartesian Product

Other Definitions

References

Introduction

Formal language

Recall that in ZFC the concepts of **pure set** and **membership relation** are primitive (undefined). Also recall that the **universe of discourse** of the theory is the **pure sets**.

Introduction

Formal language

Recall that in ZFC the concepts of **pure set** and **membership relation** are primitive (undefined). Also recall that the **universe of discourse** of the theory is the **pure sets**.

Classification

- Axioms stating the existence of sets
- Axioms determining properties of sets
- Axioms for building sets from other sets

Outline

Introduction

The Axiom of Separation

The Axiom of Power Set

The Axiom of Union

Intersection of Sets

Cartesian Product

Other Definitions

References

The Axiom of Separation

Axiom of Separation

$$\forall x \exists y \forall z (z \in y \leftrightarrow (z \in x \wedge \phi(z))),$$

where $\phi(z)$ is a formula (possibly referring to named sets (possibly with parameters)) with free variable z .

The Axiom of Separation

Axiom of Separation

$$\forall x \exists y \forall z (z \in y \leftrightarrow (z \in x \wedge \phi(z))),$$

where $\phi(z)$ is a formula (possibly referring to named sets (possibly with parameters)) with free variable z .

Informal description

"For any set x there is a set consisting of all z in x for which $\phi(z)$ holds." (p. 82)

Category

Axiom for building a set from other set.

The Axiom of Separation

Example

Let $\phi(x): x \notin x$. The Axiom of Separation postulates that

for any set x , there is a set y such that for any set z , $z \in y$ if and only if $z \in x$ and $x \notin x$ (the set y is the empty set).

The Axiom of Separation

Example

Let A and B two given sets and let

$$\phi(z, B): z \in B$$

the formula with free variable z referring to the named set B (with parameter B).

The Axiom of Separation

Example

Let A and B two given sets and let

$$\phi(z, B): z \in B$$

the formula with free variable z referring to the named set B (with parameter B).

For using the Axiom of Separation on A , B and $\phi(z, B)$ we prove that there is a set y such that for any set z , $z \in y$ if and only if $z \in A$ and $z \in B$ (the set y is the intersection of A and B).

The Axiom of Separation

Example

Let A and B two given sets and let

$$\phi(z, B): z \in B$$

the formula with free variable z referring to the named set B (with parameter B).

For using the Axiom of Separation on A , B and $\phi(z, B)$ we prove that there is a set y such that for any set z , $z \in y$ if and only if $z \in A$ and $z \in B$ (the set y is the intersection of A and B).

Question

Can you define the union of A and B using something similar to the previous example?

The Axiom of Separation

Theorem

The set postulated by the Axiom of Separation is unique.

The Axiom of Separation

Theorem

The set postulated by the Axiom of Separation is unique.

Proof

The Axiom of Separation states that for any set x , there is one set y , such that for any set z , $z \in y$ if and only if $z \in x$ and $\phi(z)$.

Let y' be other set such that for any set z , $z \in y'$ if and only if $z \in x$ and $\phi(z)$. Then $z \in y$ if and only if $z \in y'$. Therefore, $y \doteq y'$ by the Axiom of Extensionality.

The Axiom of Separation

Definition (p. 82)

Given the **existence** and **uniqueness** of the set postulated by the Axiom of Separation we can give the following definition:

Let x be a set. For each propositional function $\phi(z)$ (possibly referring to named sets (possibly with parameters (possibly with parameters))) we denote by

$$\{z \in x : \phi(z)\}$$

the set y in the Axiom of Separation such that

$$\forall z (z \in y \leftrightarrow (z \in x \wedge \phi(z))).$$

The Axiom of Separation

Remark

Note that the Axiom of Separation is a **restriction** of the Naive Comprehension Principle.

$\{z : \phi(z)\}$ (Naive Comprehension Principle),

$\{z \in x : \phi(z)\}$ (Axiom of Separation).

“The [Axiom of Separation] is thus saying that, given a set x , there are certain sets definable as subsets of x .” (p. 83)

The Axiom of Separation

Historical remark

Zermelo ([1908] 1967, p. 202) stated the Axiom of Separation using a set and a “definite property”, where the elements of the set that satisfy the definite property form a set. However, the concept of “definite property” was not accurate enough. Fraenkel ([1922] 1967) reformulated the Axiom of Separation using other concept. Skolem ([1922] 1967, pp. 202–203) defined the concept of “definite property” as it is used today, that is, it is propositional function in the formal language of ZFC. A definition similar to Skolem’s was proposed by Weyl ([1910] 2018).[†]

[†]See the introductory note by Jean van Heijenoort to (Fraenkel [1922] 1967) and (Feferman 2000).

The Axiom of Separation

Technical remark

Since there is an instance of the Axiom of Separation for each propositional function $\phi(x)$ some authors define it as an axiomatic schema.[†]

[†]See, e.g., (Enderton 1977; Hrbacek and Jech [1978] 1999; Button 2021).

Outline

Introduction

The Axiom of Separation

The Axiom of Power Set

The Axiom of Union

Intersection of Sets

Cartesian Product

Other Definitions

References

The Axiom of Power Set

Definition

A set x is a **subset** of a set y , denoted $x \subseteq y$, if every element of x belongs to y , that is,

$$x \subseteq y := \forall w (w \in x \rightarrow w \in y).$$

The Axiom of Power Set

Axiom of Power Set

$$\forall x \exists y \forall z (z \in y \leftrightarrow z \subseteq x).$$

The Axiom of Power Set

Axiom of Power Set

$$\forall x \exists y \forall z (z \in y \leftrightarrow z \subseteq x).$$

Informal description

“For any set x there is a set consisting of all subsets of x .” (p. 82)

Category

Axiom for building a set from other sets.

Remark

“Whether [this axiom] represents an intuitive statement about infinite sets is debatable. But as it leads to some of the most exciting parts of Cantor’s original work on infinite numbers, there is not much point in arguing about its inclusion!” (p. 83)

The Axiom of Power Set

Theorem

The set postulated by the Axiom of Power Set is unique.

The Axiom of Power Set

Definition (p. 82)

Given the **existence** and **uniqueness** of the set postulated by the Axiom of Power Set we can give the following definition:

Let x be a set. The **power set** of x , denoted $\mathcal{P}(x)$, is the set y in the Axiom of Power Set such that

$$\forall z (z \in y \leftrightarrow z \subseteq x).$$

That is, $\mathcal{P}(x)$ is the set of all subsets of x .

Outline

Introduction

The Axiom of Separation

The Axiom of Power Set

The Axiom of Union

Intersection of Sets

Cartesian Product

Other Definitions

References

The Axiom of Union

Axiom of Union

$$\forall x \exists y \forall z (z \in y \leftrightarrow \exists w (z \in w \wedge w \in x)).$$

The Axiom of Union

Axiom of Union

$\forall x \exists y \forall z (z \in y \leftrightarrow \exists w (z \in w \wedge w \in x)).$

Informal description

“For any set x there is a set which is the union of all the elements of x .” (p. 82)

Category

Axiom for building a set from other sets.

The Axiom of Union

Theorem

The set postulated by the Axiom of Union is unique.

The Axiom of Union

Definition (p. 82)

Given the **existence** and **uniqueness** of the set postulated by the Axiom of Union we can give the following definition:

Let x be a set. The (**generalised**) **union** of x , denoted $\bigcup x$, is the set y in the Axiom of Union such as

$$\forall z (z \in y \leftrightarrow \exists w (z \in w \wedge w \in x)).$$

The Axiom of Union

Example

(i) Let x, y be sets. Then, $z \in \bigcup\{x, y\}$ if and only if $z \in x$ or $z \in y$ (see Exercise 4.16).

The Axiom of Union

Example

- (i) Let x, y be sets. Then, $z \in \bigcup\{x, y\}$ if and only if $z \in x$ or $z \in y$ (see Exercise 4.16).
- (ii) $\bigcup \emptyset \doteq \emptyset$.

The Axiom of Union

Example

- (i) Let x, y be sets. Then, $z \in \bigcup\{x, y\}$ if and only if $z \in x$ or $z \in y$ (see Exercise 4.16).
- (ii) $\bigcup \emptyset \doteq \emptyset$.

Definition (p. 83)

Let x, y be sets. The **union** of x and y , denoted $x \cup y$, is the set $\bigcup\{x, y\}$, that is,

$$x \cup y := \bigcup\{x, y\}.$$

The Axiom of Union

Exercise

Let x_1, x_2, x_3 be sets. Show that there is a set y such that $z \in y$ if and only if $z \in x_1$ or $z \in x_2$ or $z \in x_3$.

The Axiom of Union

Exercise

Let x_1, x_2, x_3 be sets. Show that there is a set y such that $z \in y$ if and only if $z \doteq x_1$ or $z \doteq x_2$ or $z \doteq x_3$.

Solution

By the Axiom of Pairs $\{x_1, x_2\}$ and $\{x_3\}$ are sets. Now, by the definition of union of sets and the Axiom of Union, we can define the set y by

$$\{x_1, x_2\} \cup \{x_3\} \doteq \bigcup \{\{x_1, x_2\}, \{x_3\}\}.$$

The Axiom of Union

Exercise

Let x_1, x_2, x_3 be sets. Show that there is a set y such that $z \in y$ if and only if $z \doteq x_1$ or $z \doteq x_2$ or $z \doteq x_3$.

Solution

By the Axiom of Pairs $\{x_1, x_2\}$ and $\{x_3\}$ are sets. Now, by the definition of union of sets and the Axiom of Union, we can define the set y by

$$\{x_1, x_2\} \cup \{x_3\} \doteq \bigcup \{\{x_1, x_2\}, \{x_3\}\}.$$

Notation

From the previous exercise we extended the curly bracket notation denoting by $\{x_1, x_2, x_3\}$ the set whose elements are x_1, x_2, x_3 . In general, $\{x_1, x_2, \dots, x_n\}$ denotes the set with $x_1, x_2, \dots, x_n$ as its elements.

Outline

Introduction

The Axiom of Separation

The Axiom of Power Set

The Axiom of Union

Intersection of Sets

Cartesian Product

Other Definitions

References

Intersection of Sets

Definition (p. 85)

Let x, y be sets. The **intersection** of x and y , denoted $x \cap y$, is defined by

$x \cap y := \{z : (z \in x \wedge z \in y)\},$ Wrong! (Naive Comprehension Principle),

$x \cap y := \{z \in x \cup y : (z \in x \wedge z \in y)\},$ Right! (Axiom of Separation).

Intersection of Sets

Definition (Exercise 4.22)

Let x be a set. The **(generalised) intersection** of x , denoted $\bigcap x$, is defined by

$$\bigcap x := \{z : z \in y \text{ for all } y \in x\},$$

Wrong! (Naive Comprehension Principle),

Intersection of Sets

Definition (Exercise 4.22)

Let x be a set. The **(generalised) intersection** of x , denoted $\bigcap x$, is defined by

$$\bigcap x := \{z : z \in y \text{ for all } y \in x\}, \quad \text{Wrong! (Naive Comprehension Principle),}$$

$$\bigcap x := \{z \in \bigcup x : z \in y \text{ for all } y \in x\}, \quad \text{Right! (Axiom of Separation).}$$

Outline

Introduction

The Axiom of Separation

The Axiom of Power Set

The Axiom of Union

Intersection of Sets

Cartesian Product

Other Definitions

References

Cartesian Product

Definition (p. 86)

Let X, Y be sets. The **Cartesian Product** of X and Y , denoted $X \times Y$, is defined by

$$X \times Y := \{ \langle x, y \rangle : (x \in X \wedge y \in Y) \},$$

Wrong! (Naive Comprehension Principle),

Cartesian Product

Definition (p. 86)

Let X, Y be sets. The **Cartesian Product** of X and Y , denoted $X \times Y$, is defined by

$$X \times Y := \{ \langle x, y \rangle : (x \in X \wedge y \in Y) \},$$

Wrong! (Naive Comprehension Principle),

$$X \times Y := \{ z : \exists x \exists y ((x \in X \wedge y \in Y) \wedge z \doteq \langle x, y \rangle) \},$$

Still wrong! (Naive Comprehension Principle),

Cartesian Product

Definition (p. 86)

Let X, Y be sets. The **Cartesian Product** of X and Y , denoted $X \times Y$, is defined by

$$X \times Y := \{ \langle x, y \rangle : (x \in X \wedge y \in Y) \},$$

Wrong! (Naive Comprehension Principle),

$$X \times Y := \{ z : \exists x \exists y ((x \in X \wedge y \in Y) \wedge z \doteq \langle x, y \rangle) \},$$

Still wrong! (Naive Comprehension Principle),

$$X \times Y := \{ z \in \mathcal{P}(\mathcal{P}(X \cup Y)) : \exists x \exists y ((x \in X \wedge y \in Y) \wedge z \doteq \langle x, y \rangle) \},$$

Right! (Axiom of Separation).

(continued on next slide)

Cartesian Product

Definition (p. 86, continuation)

Note that $z \doteq \{\{x\}, \{x, y\}\}$.

Cartesian Product

Definition (p. 86, continuation)

Note that $z \doteq \{\{x\}, \{x, y\}\}$.

Since $x \in X$ and $y \in Y$ then $x, y \in X \cup Y$.

Cartesian Product

Definition (p. 86, continuation)

Note that $z \doteq \{\{x\}, \{x, y\}\}$.

Since $x \in X$ and $y \in Y$ then $x, y \in X \cup Y$.

Also note that $\{x, y\} \in \mathcal{P}(X \cup Y)$ and $\{x\} \in \mathcal{P}(X \cup Y)$.

Cartesian Product

Definition (p. 86, continuation)

Note that $z \doteq \{\{x\}, \{x, y\}\}$.

Since $x \in X$ and $y \in Y$ then $x, y \in X \cup Y$.

Also note that $\{x, y\} \in \mathcal{P}(X \cup Y)$ and $\{x\} \in \mathcal{P}(X \cup Y)$.

Therefore $\{\{x\}, \{x, y\}\} \in \mathcal{P}(\mathcal{P}(X \cup Y))$.

Outline

Introduction

The Axiom of Separation

The Axiom of Power Set

The Axiom of Union

Intersection of Sets

Cartesian Product

Other Definitions

References

Other Definitions

Let x and y be sets. From the previous six axioms, we can also define the concepts of **complement** of x in y , **function**, **domain** and **range** of a function, **restriction** and **extension** of a function, the **set of all functions** from x to y and **sequence of elements of a set**.

Outline

Introduction

The Axiom of Separation

The Axiom of Power Set

The Axiom of Union

Intersection of Sets

Cartesian Product

Other Definitions

References

References

- Tim Button (2021). Set Theory. An Open Introduction. Fall 2021. Revision 2138a50 (2026-07-11). Independently published. URL: <https://builds.openlogicproject.org/courses/set-theory/> (cit. on p. 19).
- Herbert B. Enderton (1977). Elements of Set Theory. Academic Press (cit. on p. 19).
- Solomon Feferman (2000). The Significance of Weyl's *Das Kontinuum*. History and Philosophical Significance. In: Proof Theory, History and Philosophical Significance. Ed. by Vincent F. Hendricks, Stig Andur Pedersen and Klaus Frovin Jørgensen. Vol. 292. Synthese Library. Studies in Epistemology, Logic, Methodology, and Philosophy of Science. Kluwer Academic Publishers, pp. 179–194. DOI: [10.1007/978-94-017-2796-9_8](https://doi.org/10.1007/978-94-017-2796-9_8) (cit. on p. 18).
- Abraham A. Fraenkel (1922). Über den Begriff “definit” und die Unabhängigkeit des Auswahlaxioms. In: Sitzungsbericht der Preussischen Akademie der Wissenschaften, Physikalisch-mathematische Klasse, pp. 253–257 (cit. on p. 52).
- [1922] (1967). The Notion “Definite” and the Independence of the Axiom of Choice. In: From Frege to Gödel: A Source Book in Mathematical Logic, 1879–1931. Ed. by Jean van Heijenoort. Trans. by Beverly Woodward. Source Books in the History of the Sciences. Translation of Fraenkel (1922). Harvard University Press, pp. 284–289 (cit. on p. 18).

References

- Derek Goldrei (1996). *Classic Set Theory. A Guided Independent Study*. Chapman & Hall (cit. on p. 2).
- Karel Hrbacek and Thomas Jech [1978] (1999). *Introduction to Set Theory*. Third Edition, Revised and Expanded. Marcel Dekker (cit. on p. 19).
- Toralf Skolem (1922). Einige Bemerkungen zur axiomatischen Begründung der Mengenlehre. In: *Proceedings of the the 5th Scand. Math. Congress. Helsinki*, pp. 217–232 (cit. on p. 53).
- [1922] (1967). Some Remarks on Axiomatized Set Theory. In: *From Frege to Gödel: A Source Book in Mathematical Logic, 1879–1931*. Ed. by Jean van Heijenoort. Trans. by Stefan Bauer-Mengelberg. *Source Books in the History of the Sciences*. Translation of Skolem (1922). Harvard University Press, pp. 290–301 (cit. on p. 18).
- Jean van Heijenoort, ed. (1967). *From Frege to Gödel: A Source Book in Mathematical Logic, 1879–1931*. *Source Books in the History of the Sciences*. Harvard University Press.
- Hermann Weyl (1910). Über die Definitionen der mathematischen Grundbegriffe, *Mathematisch-naturwissenschaftliche Blätter* 7, pp. 93–95, 109–113 (cit. on p. 53).
- [1910] (2018). *Definitions of Fundamental Mathematical Concepts*. Trans. by Stephen Pollard. Translation of Weyl (1910). URL: https://www.academia.edu/36559178/Definitions_of_Fundamental_Mathematical_Concepts (cit. on p. 18).

References

- E. Zermelo (1908). Untersuchungen über die Grundlagen der Mengenlehre. I. *Mathematische Annalen* 65.2, pp. 261–281. DOI: [10.1007/BF01449999](https://doi.org/10.1007/BF01449999) (cit. on p. 54).
- Ernst Zermelo [1908] (1967). Investigations in the Foundations of Set Theory I. In: *From Frege to Gödel: A Source Book in Mathematical Logic, 1879–1931*. Ed. by Jean van Heijenoort. Trans. by Stefan Bauer-Mengelberg. *Source Books in the History of the Sciences*. Translation of Zermelo (1908). Harvard University Press, pp. 199–215 (cit. on p. 18).